

Documentation for Authority Link Correction Scripts

Project Scope & Purpose

This suite of Python scripts (`replace_wikidata.ipynb`, `replace_lcnaf.ipynb`, `replace_geo.ipynb` and `replace_viaf.ipynb`) is developed as an analytical, verification, and audit pipeline to assist the project team in reconciling, replacing, and enriching authority linkages across the dataset.

Key Objectives:

- **Data Audit & Verification:** To programmatically scan, replace outdated links, and validate existing authority anchors (Wikidata, LCNAF, GeoNames, VIAF) against baseline metadata.
- **Missing Link Insertion & Reconciliation:** To automatically identify missing authority links for creators, receivers, and geographic locations using multi-attribute lookup and fuzzy-matching logic.
- **Decision Support & Quality Control:** To generate detailed log sheets (e.g., replacement logs, insertion logs, ambiguity reports) that provide transparent, actionable feedback for catalogers and team review.
- **Workflow Reference:** To establish a testable prototype for linked data normalization, serving as a functional reference and benchmark for the project team's final metadata compilation.

Note: Output datasets and log sheets generated by these scripts serve as benchmark verification reports and candidate datasets for team review rather than the final published release.

1. Script 1: Wikidata Label and URI Replacement

This script updates existing Wikidata anchors in the linked columns. It now performs two actions: it replaces the visible label when a corrected label is available, and it normalizes the hyperlink to the Wikidata entity URI.

Input and Output Files

```
MAIN_FILE = export_stars_saa_wikidata_replaced_v1.xlsx
CORRECTIONS_FILE = Wikidata_person_updated.xlsx
OUTPUT_FILE = export_stars_saa_wikidata_replaced_v2.xlsx
```

Target Columns

```
creator_linked_new
receiver_linked_new
sender_place_linked_new
receiver_place_linked_new
```

Required Correction File Columns

```
[in Wikidata_person_updated.xlsx]
LODwikiData
Q_ID
wikidata_label_final
```

Processing Logic

The script extracts the QID from either a Wikidata browser URL, a Wikidata entity URI, or a standalone QID. It uses the QID to look up the corrected label and then rewrites the anchor to the canonical entity URI.

Example Correction

Old:

```
<a href="https://www.wikidata.org/wiki/Q310439">Fox, George</a>      ??
```

New:

```
<a href="http://www.wikidata.org/entity/Q310439">George Fox (Q310439)</a>
```

The URI is changed from a browser page to the entity URI, and the label is changed to the Wikidata-style display label with the QID included.

Log Sheet

The script creates `wikidata_replacement_log`. Review this sheet to confirm the row, column, QID, old href, new href, old label, and new label. Rows in this sheet are the changes that the script made.

2. Script 2: LCNAF Replacement and Insertion

This script updates LCNAF authority links and inserts missing LCNAF links for creators and receivers when a safe name match is found.

Input and Output Files

```
EXPORT_FILE = export_stars_saa_wikidata_replaced_v2.xlsx
CORRECTED_LCNAF_FILE = LCNAF_updated.xlsx
OUTPUT_FILE = ssa_lcnaf_updated.xlsx
```

Target Columns

```
creator_linked_new
receiver_linked_new
sender_place_linked_new
receiver_place_linked_new
```

Required Correction File Columns

```
[in LCNAF_updated.xlsx]
personStdName
LODLOC_inDB or LODLOC
LODLOC_uri
LODLOC_label
```

Processing Logic

The script first replaces existing LCNAF anchors by matching old or corrected LCNAF IDs. It then inserts missing LCNAF links into `creator_linked_new` and `receiver_linked_new` using author name fields and receiver values.

Name Matching and Life Dates

The script removes life dates only for matching. This allows a value such as Harrison, James, 1628-1687 to match James Harrison or Harrison, James. For this project, first-name and last-name matching is sufficient when it returns one unique record.

If more than one LCNAF record matches the same normalized name, such as possible father/son cases with the same first and last name but different years, the script does not choose automatically. It skips the insertion and writes the case to the ambiguity log for manual review.

Example Correction

Old:

```
<a href="https://id.loc.gov/authorities/names/n79065610.html">Fox, George</a>
```

New:

```
<a href="http://id.loc.gov/authorities/names/n79065610">Fox, George, 1624-1691</a>
```

Log Sheets and Review Instructions

Log sheet	What it means
lcnaf_replace_log	Existing LCNAF links were replaced.
lcnaf_insert_log	New LCNAF links were inserted by name match.
lcnaf_ambiguous_insert_log	More than one possible LCNAF match was found.
lcnaf_existing_missing	Existing LCNAF link was not in the correction file.
lcnaf_duplicate_old_ids	Duplicate old LCNAF IDs appeared in the correction file.
lcnaf_not_applied	Correction record was not used.

3. Script 3: GeoNames Replacement and Insertion

This script updates GeoNames links and inserts missing GeoNames links for sender and receiver places.

Input and Output Files

```
EXPORT_FILE = ssa_lcnaf_updated.xlsx
CORRECTED_GEO_FILE = GeoNames_updated.xlsx
OUTPUT_FILE = ssa_geo_updated.xlsx
```

Target Columns

Replacement columns:

```
creator_linked_new
receiver_linked_new
sender_place_linked_new
receiver_place_linked_new
```

Insertion columns:

```
sender_place_linked_new
receiver_place_linked_new
```

Processing Logic

The script replaces existing GeoNames anchors when an old GeoNames ID maps to a corrected record. It also inserts missing GeoNames links using sender_place and receiver_place values. Matching begins with exact normalized matching and then uses fuzzy matching with stricter thresholds for one-word places.

GeoNames Example

Subject place:

Lancaster Castle, Lancaster, Lancashire, England

Updated GeoNames label:

Lancaster Castle (Lancaster, Lancashire, England)

Corrected URI:

[https://sws.geonames.org/\[GeoNames_ID\]](https://sws.geonames.org/[GeoNames_ID])

The script moves from the main-file wording to the corrected GeoNames label and uses the canonical sws.geonames.org URI.

Ambiguity Resolver

When multiple fuzzy GeoNames matches are close, the script compares the subject place before the first comma with the corrected GeoNames label before the first bracket. If exactly one record matches that prefix, the script uses it. Otherwise, the case is logged as ambiguous and skipped.

Log Sheets and Review Instructions

Log sheet	What it means
geo_replace_log	Existing GeoNames links were replaced.
geo_insert_log	New GeoNames links were inserted.
geo_ambig_log	Multiple close GeoNames matches were found.
geo_not_in_corr	Existing GeoNames link was not in the correction file.
geo_duplicate_ids	Duplicate old GeoNames IDs appeared.
geo_no_place_match	No safe match was found for a place value.
geo_not_applied	Correction record was not used.

4. Script 4: VIAF Replacement and Insertion

This script updates VIAF authority links and inserts missing VIAF links for creators and receivers.

Input and Output Files

```
EXPORT_FILE = ssa_geo_updated.xlsx
PERSON_FILE = VIAF_updated.xlsx
OUTPUT_FILE = ssa_viaf_updated.xlsx
```

Processing Logic

The script replaces existing VIAF anchors when the old or corrected VIAF ID appears in the correction file. It inserts missing VIAF links by matching creator and receiver names. Like LCNAF, life dates are removed only for matching, and ambiguous same-name cases are skipped and logged.

Example Correction

Old:

Fox, George

New:

Fox, George, 1624-1691

A Full Example showing replacement and insertion:

Old:

creator_linked

<p>Margaret Fell
[LC]</p><p>Margaret Fell (Q524860)
[Wikidata]</p>

Display:

[Margaret Fell](https://id.loc.gov/authorities/names/n79100544) [LC]

[Margaret Fell \(Q524860\)](https://www.wikidata.org/wiki/Q524860) [Wikidata]

New:

<p>Fell, Margaret, 1614-1702
[LC]</p><p>Margaret Fell (Q524860)
[Wikidata]</p><p>Fell, Margaret, 1614-1702
[VIAF]</p>

Display:

[Fell, Margaret, 1614-1702](http://id.loc.gov/authorities/names/n79100544) [LC]

[Margaret Fell \(Q524860\)](https://www.wikidata.org/wiki/Q524860) [Wikidata]

[Fell, Margaret, 1614-1702](https://viaf.org/viaf/7421776) [VIAF]

Old:

Receiver_linked:

<p>Thomas Fell (1598-1658)
[LC]</p><p>Thomas Fell (1598-1658)
(Q7789545) [Wikidata]</p>

[Thomas Fell \(1598-1658\)](https://id.loc.gov/authorities/names/nr95011120) [LC]

[Thomas Fell \(1598-1658\) \(Q7789545\)](https://www.wikidata.org/wiki/Q7789545) [Wikidata]

New:

<p>Fell, Thomas, 1598-1658
[LC]</p><p>Thomas Fell (Q7789545)
[Wikidata]</p><p>Fell, Thomas, 1598-1658
[VIAF]</p>

Receiver [Linked data]

[Fell, Thomas, 1598-1658](http://id.loc.gov/authorities/names/nr95011120) [LC]

[Thomas Fell \(Q7789545\)](https://www.wikidata.org/wiki/Q7789545) [Wikidata]

[Fell, Thomas, 1598-1658](https://viaf.org/viaf/26952118) [VIAF]

Log Sheets and Review Instructions

Log sheet	What it means
viaf_replace_log	Existing VIAF links were replaced.
viaf_insert_log	New VIAF links were inserted.
viaf_ambig_log	More than one VIAF match was possible.
viaf_existing_missing	Existing VIAF link was not in the

	correction file.
viaf duplicate ids	Duplicate old VIAF IDs appeared.
viaf not applied	Correction record was not used.

8. Required Packages

```
!pip install -q openpyxl tqdm rapidfuzz
```

Project Team

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